Computational Statistics
Unit 2: Random Variables

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Discrete random variables

Our first aim is to simulate discrete or continuous random variables in $\mathbb{R}$.

Discrete random variable

A discrete random variable can take a finite or countable number of values. Given a random variable X, its probability mass function is given by

$$f(x) = \mathbb{P}(X = x)$$

In general, to describe the distribution of discrete random variable we just need to identify real values $p[1], p[2], \ldots, p[n], \ldots$ such that

1. $p[i] \in (0, 1]$
2. $\sum_{i=1}^{\infty} p[i] = 1$

and real numbers $x[1], \ldots, x[n], \ldots$, such that

$$\mathbb{P}(X = x[i]) = p[i]$$

In general, the vector $p = (p[1], \ldots, p[n], \ldots)$ is called a probability vector.

Discrete random variables

Bernoulli random variable

A random variable X is said to have Bernoulli distribution of parameter p, written $X \sim \text{Ber}(p)$ if

$$\mathbb{P}(X = 1) = p \quad \text{and} \quad \mathbb{P}(X = 0) = 1 - p.$$

Discrete Uniform

A random variable X is said to have discrete uniform distribution of parameter n if X takes values in $\{1, 2, \ldots, n\}$ and

$$\mathbb{P}(X = i) = 1/n \quad \text{for all } i \in \{1, \ldots, n\}$$

Binomial Distribution

A random variable X is said to have Binomial distribution of parameters n and p, denoted $X \sim \mathrm{Bin}(n, p)$, if its probability mass function is given by

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq k \leq n$$

Exercise: Let $X_1, \ldots, X_n$ be independent and identically distributed $\mathrm{Ber}(p)$ random variables. Show that $X = X_1 + \ldots + X_n$ has binomial distribution of parameters n and p.

Poisson Distribution

A random variable X is said to have Poisson distribution of parameters λ, denoted $X \sim \text{Poi}(\lambda)$, if its probability mass function is given by

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2 \ldots$$

Geometric Distribution

A random variable X is said to have Geometric distribution of parameters p, denoted $X \sim \text{Geo}(p)$, if its probability mass function is given by

$$\mathbb{P}(X = k) = p(1-p)^{k-1}, \quad k = 1, 2, 3 \ldots$$

Exercise: Let $(X_i)_{i=1}^{\infty}$ be a sequence of independent $\text{Ber}(p)$ random variables. Show that $T = \min\{i \geq 1 : X_i = 1\}$ has $\text{Geo}(p)$ distribution.

Mean and Variance

Definition: mean

We define the mean of a discrete random variable X taking values in $\{x_1, x_2, \ldots\}$ as

$$\mathbb{E}(X) = \sum_{j=1}^{\infty} x_j p_j$$

where $p_j = \mathbb{P}(X = x_j)$

Definition: moments

We define the k-th moment of a discrete random variable X taking values in $\{x_1, x_2, \ldots\}$ as

$$\mathbb{E}(X^k) = \sum_{j=1}^{\infty} x_j^k p_j.$$

Definition: variance

The variance of a random variable X is given by $\mathbb{V}ar(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$

1. Continuous Random Variables

Continuous Random Variables in $\mathbb{R}$

Let's recall a bit what are continuous random variables

Continuous random variables

A continuous real random variable takes values in an interval or a disjoint union of them[a].

Given a real random variable X, its cumulative distribution function is given by

$$F(x) = \mathbb{P}(X \leq x)$$

For **continuous** random variables $F(x)$ is differentiable. $f(x) = \frac{d}{dx}F(x)$ is called the density function of X

[a] A formal definition requires notions of Measure Theory

Continuous Random Variables in $\mathbb{R}$

In general we will describe continuous random variables in terms of its density function or its cumulative distribution function.

Density functions: any function $f : \mathbb{R} \to \mathbb{R}$ satisfying

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

is a density function of a continuous random variable.

Cumulative distribution functions: any function $F : \mathbb{R} \to \mathbb{R}$ satisfying

1. F is increasing
2. $\lim_{x \to -\infty} F(x) = 0$
3. $\lim_{x \to \infty} F(x) = 1$

is the cumulative distribution function of a random variable. If additionally

4. F is differentiable[1]

Then F is the **cdf** of a **continuous** random variable

[1] technically, differentiable almost everywhere

Mean and Variance

For a continuous random variable we can easily compute its expectation and moments in general.

Expectation

Let X be a continuous random variable with density f, and let $\phi : \mathbb{R} \to \mathbb{R}$ be a function. Then $\phi(X)$ is also a random variable, and moreover

$$\mathbb{E}(\phi(X)) = \int_{-\infty}^{\infty} \phi(x) f(x) dx.$$

In particular, $\mathbb{E}(X) = \int x f(x) dx$ and $\mathbb{E}(X^2) = \int x^2 f(x) dx$

Recall that $\mathbb{V}ar(X) = \mathbb{E}\left((X - \mathbb{E}(X))^2\right) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$

Uniform Distribution

The **MOST** important distribution.

Uniform Distribution

U is said to have uniform distribution in $[a, b]$ if its density function f is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases},$$

or simply $f(x) = \frac{1}{b-a}\mathbb{1}_{[a,b]}(x)$

As we already discuss, U is a primitive for simulation algorithms, and we usually generate pseudo-random numbers trying to imitate a uniform distribution.

From now on, if U has uniform distribution in $[0, 1]$ we simply write $U \sim \mathcal{U}$

Recall that in Python, particularly using nympy, the function `np.random.uniform(m)` gives m independent samples of a uniform distribution[2] and `np.random.uniform(m,a,b)` returns m samples of uniform random variables in the interval $[a, b]$

[2] technically, pseudo-random

Important Distributions

Important continuous distributions in $\mathbb{R}$

1. **Exponential:** $\text{Exp}(\lambda)$**,** $\lambda > 0$**.**

$$f(x) = \lambda e^{-\lambda x} \mathbb{1}_{[0,\infty)}(x)$$

2. **Gamma:** $\text{Gamma}(\alpha, \beta)$, $\alpha > 0, \beta > 0$**:**

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

3. **Beta:** $\text{Beta}(\alpha, \beta)$, $\alpha > 0, \beta > 0$.

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \mathbb{1}_{[0,1]}(x)$$

4. **Normal**: $\mathcal{N}(\mu, \sigma)$, $\mu \in \mathbb{R}$, $\sigma > 0$.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

To get *m* sample of the above distributions we can use `np.random.exp(1/l,m)`, `np.random.gamma(a, 1/b,m)`, `np.random.beta(a,b,m)`
`np.random.normal(mu,sigma,m)`

Multivariate distributions

A random variable X taking values in $\mathbb{R}^d$ with $d \geq 2$ is called a multivariate random variable or random vector

A multivariate random variable X is usually characterised by its density function $f : \mathbb{R}^d \to \mathbb{R}$ which satisfies the following properties:

1. $f(x) \geq 0$ for all $x \in \mathbb{R}^d$
2. $\int_{\mathbb{R}^d} f(x)dx = 1$
3. $\mathbb{P}(X \in A) = \int_A f(x)dx$

Multivariate Normal Distribution

In general, most people remember between 0 to 1 multivariate distribution. If they remember one, it is likely the multivariate normal distribution.

(non-degenerate) Multivariate Normal Distribution

A random vector X taking values in $\mathbb{R}^d$ is said to have multivariate normal distribution with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$, in short $X \sim \mathcal{N}(\mu, \Sigma)$, if its density function $f : \mathbb{R}^d \to \mathbb{R}$ is

$$f(x) = \frac{1}{(2\pi)^{d/2}\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2}(x-\mu)^\intercal \Sigma^{-1}(x-\mu)\right).$$

Not any matrix Σ can be be a covariance matrix. A valid covariance matrix satisfies:

1. Σ is symmetric
2. Σ is definite positive:

$$\text{for all } x \in \mathbb{R}^d \setminus \{\mathbf{0}\} \text{ it holds that } x^\intercal \Sigma x > 0$$

or equivalently

$$\text{All the eigenvalues of } \Sigma \text{ are positive}$$

Multivariate Normal Distribution

There are two important properties of (multivariate) normal distributions that are worth mentioning:

Proposition

If we generate $(Z_i)_{i=1}^d \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$*, then the vector* $Z = (Z_1, \ldots, Z_n) \sim \mathcal{N}(0, I)$ *where* I *is the* $d \times d$ *identity*

Theorem

$X \sim \mathcal{N}(\mu, \Sigma)$ *if and only if it exists* $Z \sim \mathcal{N}(0, I)$ *such that* $X = \mu + AZ$ *with* $AA^\intercal = \Sigma$

To sample from a multivariate normal we can use `np.random.multivariate_normal`